

Faezeh Khazaei

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SUMMARY

PhD applicant in machine learning for computational biology, seeking a PhD position in ML for single-cell genomics. Experienced in deep learning, self-supervised and representation learning, and generative modeling on large-scale single-cell and genomic data — spanning interpretable scRNA-seq phenotype prediction, self-supervised transformers on regulatory DNA, and integrating chromatin accessibility and histone-modification signals to model cell-type-specific gene regulation.

EDUCATION

M.Sc., Computer Science — University of California, Davis (GPA 4.0/4.0) 2023 – Dec 2026 (expected)
M.Sc., Information Technology Engineering — University of Tehran (GPA 18.5/20) 2020 – 2023
B.Sc., Information Technology Engineering — Sharif University of Technology (GPA 17.25/20) 2015 – 2019

RESEARCH EXPERIENCE

Graduate Researcher — Quon Lab, University of California, Davis Oct 2023 – Present

- **CELECTION:** interpretable multiple-instance learning for phenotype prediction from single-cell RNA-seq; led benchmarking, validation, figure design, and manuscript preparation.
- **Promoters2Gene (P2G):** predicting cell-type-specific gene expression by integrating DNA sequence with chromatin accessibility (ATAC-seq) and histone-modification (ChIP-seq) signals.
- **ProMod:** self-supervised transformer on promoter DNA across 100+ vertebrates for gene-expression prediction.
- **Varformer:** set-based transformer for rare-variant disease prediction from regulatory and gene-context features.
- **EpiCRISPR:** structural and guide-activity determinants of CRISPR epigenome-editing outcomes from large-scale gRNA libraries.
- **AvalancheBreak:** how periodic boundary/separator tokens shape transformer behavior across algorithmic tasks — boundaries as scaffolding, not added capability.

Graduate Research Assistant — Robotics & Machine Intelligence Lab, University of Tehran Oct 2020 – Aug 2023

- Developed a transformer-based multi-hop co-attention network for visual question answering (VQA), improving reasoning across multimodal inputs by 10%.
- Pseudo-human non-player agents in a serious game with vision-based perception and decision constraints, from behavioral data of 50 participants.

PUBLICATIONS

- **CELECTION: Interpretable Multiple-Instance Learning for Single-Cell Phenotype Prediction.** Preprint; contributed benchmarking, validation, figures, writing, and peer-review revisions; named author on the forthcoming published version.
- **EpiCRISPR: Predicting CRISPR Epigenome-Editing Outcomes.** Preprint forthcoming (October 2026).
- **Varformer: Modeling Somatic Mosaicism in Brain Disorders.** Preprint forthcoming (October 2026).
- **AvalancheBreak: The Role of Boundary/Separator Tokens in Transformer Models.** Preprint forthcoming (September 2026).

TEACHING EXPERIENCE

Teaching Assistant, UC Davis — Introduction to Programming (ECS 32A) Summer 2026

Teaching Assistant, University of Tehran / Sharif Univ. of Technology 2018 – 2023
Neural Networks, Multimedia Systems, Information Retrieval, and Special Topics in CS — supported 50+ students.

INDUSTRY EXPERIENCE

Freelance Developer / Data Scientist Nov 2020 – May 2023

- ML pipelines, recommender systems, and analytics dashboards for 10+ real-estate and e-commerce clients (Python, SQL, deep learning); 10k+ records, 10,000+ users.

SKILLS

Programming Python, R, SQL, MATLAB, C, JavaScript, Bash

ML / Deep Learning PyTorch, JAX, TensorFlow, Triton, scikit-learn, NumPy, Pandas; transformers, self-supervised & representation learning, generative & foundation models, SFT, RLHF

Single-Cell & Genomics scRNA-seq, Scanpy, scverse, AnnData, single-cell atlases, multi-omics integration, ATAC-seq, ChIP-seq, regulatory genomics

Tools Git, SLURM/HPC, Docker, AWS, MLflow, Azure DevOps, Power BI

HONORS & AWARDS

Ranked 7th nationwide, national M.Sc. entrance exam (2020) · Top 1%, national undergraduate entrance exam (2015).